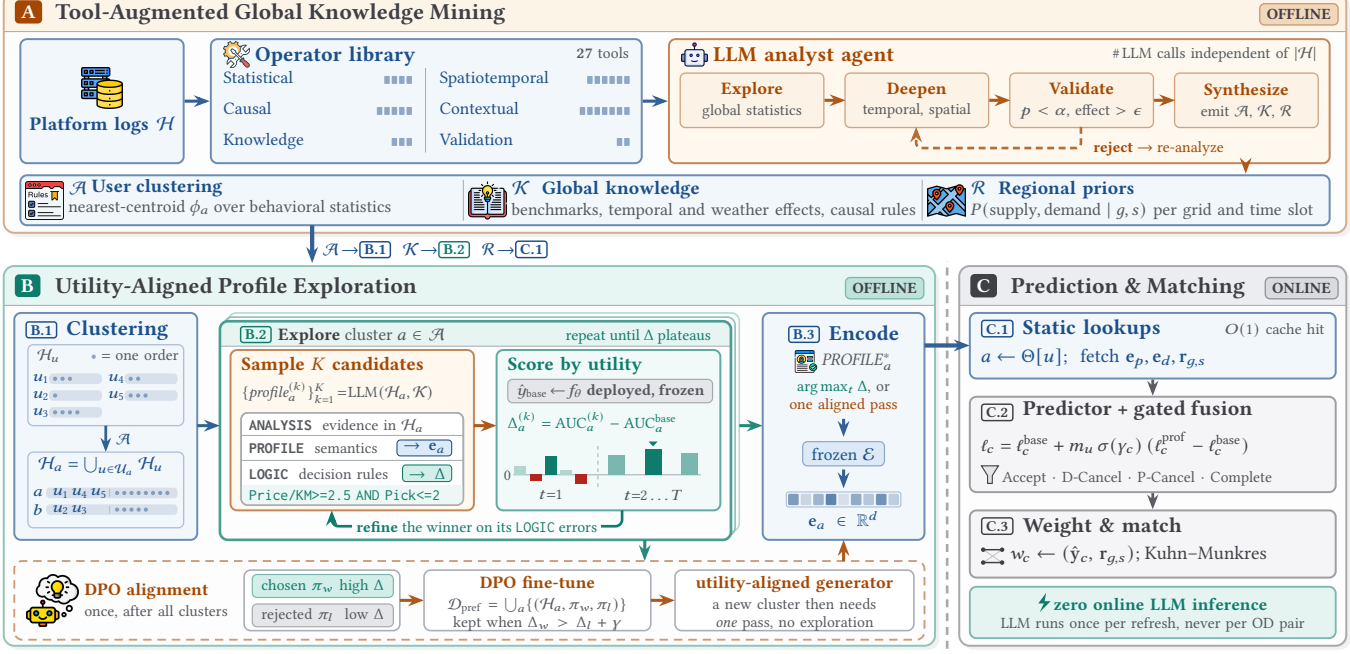

A Tool-Augmented Global Knowledge Mining
OFFLINE
Platform logs $\mathcal{H}$
Operator library
27 tools
Statistical
Causal
Knowledge
Spatiotemporal
Contextual
Validation
LLM analyst agent
LLM calls independent of $|\mathcal{H}|$
Explore
global statistics
Deepen
temporal, spatial
Validate
$p < \alpha$, effect $> \epsilon$
Synthesize
emit $\mathcal{A}, \mathcal{K}, \mathcal{R}$
reject → re-analyze
$\mathcal{A}$ User clustering
nearest-centroid ϕ_a over behavioral statistics
$\mathcal{K}$ Global knowledge
benchmarks, temporal and weather effects, causal rules
$\mathcal{R}$ Regional priors
$P(\text{supply, demand} \mid g, s)$ per grid and time slot
$\mathcal{A}$ → B.1 $\mathcal{K}$ → B.2 $\mathcal{R}$ → C.1
B Utility-Aligned Profile Exploration
OFFLINE
B.1 Clustering
$\mathcal{H}_u$ ∘ = one order
u_1 u_4
u_2 u_5
u_3
$\mathcal{A}$
$\mathcal{H}_a = \bigcup_{u \in \mathcal{U}_a} \mathcal{H}_u$
a u_1 u_4 u_5
b u_2 u_3
B.2 Explore cluster $a \in \mathcal{A}$
repeat until Δ plateaus
Sample K candidates
$\{profile_a^{(k)}\}_{k=1}^{K}$ = LLM$(\mathcal{H}_a, \mathcal{K})$
ANALYSIS evidence in $\mathcal{H}_a$
PROFILE semantics → $\mathbf{e}_a$
LOGIC decision rules → Δ
Price/KM>=2.5 AND Pick<=2
Score by utility
$\hat{y}_{\text{base}} \leftarrow f_\theta$ deployed, frozen
$\Delta_a^{(k)} = \text{AUC}_a^{(k)} - \text{AUC}_a^{\text{base}}$
0
t=1
t=2 . . . T
refine the winner on its LOGIC errors
B.3 Encode
$PROFILE_a^*$
arg max$_t$ Δ, or one aligned pass
frozen $\mathcal{E}$
$\mathbf{e}_a \in \mathbb{R}^d$
DPO alignment
once, after all clusters
chosen π_w high Δ
rejected π_l low Δ
DPO fine-tune
$\mathcal{D}_{\text{pref}} = \bigcup_a \{(\mathcal{H}_a, \pi_w, \pi_l)\}$
kept when $\Delta_w > \Delta_l + \gamma$
utility-aligned generator
a new cluster then needs one pass, no exploration
C Prediction & Matching
ONLINE
C.1 Static lookups
$O(1)$ cache hit
$a \leftarrow \Theta[u]$; fetch $\mathbf{e}_p, \mathbf{e}_d, \mathbf{r}_{g,s}$
C.2 Predictor + gated fusion
$\ell_c = \ell_c^{\text{base}} + m_u\, \sigma(\gamma_c)\, (\ell_c^{\text{prof}} - \ell_c^{\text{base}})$
Accept · D-Cancel · P-Cancel · Complete
C.3 Weight & match
$w_c \leftarrow (\hat{y}_c, \mathbf{r}_{g,s})$; Kuhn–Munkres
zero online LLM inference
LLM runs once per refresh, never per OD pair